Module 4 Assignment

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Following is the link to my GitHub account:

https://github.com/Kgardner22/64018_kgardner

LP Problem: The Weigelt Corporation

If lpSolveAPI is not already installed, remove the "*" preceeding lines 26-30 and run the block:

*Install lpSolveAPI

*{r} *install.packages("lpSolveAPI") *

load the needed library

```
library(lpSolveAPI)
```

We will setup the Weigelt Corporation problem. Note, we have nine (9) decision variables and twelve (12) constraints. We also have nine (9) non-negativity constraints.

We will read the lp formulation using an lp file.

```
x <- read.lp("Weigelt.lp") # create an lp object x
x                          # display x

## Model name:
##   a linear program with 9 decision variables and 12 constraints
```

Solve the lp model

```
solve(x)

## [1] 0

get.objective(x)        # get objective value
```

```
## [1] 696000

get.variables(x)        # get values of decision variables

## [1] 516.6667   0.0000   0.0000 177.7778 666.6667   0.0000   0.0000
166.6667
## [9] 416.6667

get.constraints(x)      # get constraint RHS values

##  [1]   694.4444   833.3333   416.6667 13000.0000 12000.0000  5000.0000
##  [7]   516.6667   844.4444   583.3333     0.0000     0.0000     0.0000
```